MCFA-UNet: Multiscale Cascaded Feature Attention U-Net for Liver Segmentation

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Abstract

Medical image segmentation has become an essential task in computer-aided diagnosis and treatment planning. Among various applications, liver segmentation from CT images is particularly important due to its role in detecting tumors and planning surgical procedures. However, accurate liver segmentation remains challenging because of low contrast boundaries, irregular shapes, and similarity with surrounding organs. This work focuses on improving liver segmentation performance using a deep learning-based approach called MCFA-UNet (Multiscale Cascaded Feature Attention U-Net). The model enhances the traditional U-Net architecture by incorporating multiscale feature extraction to capture contextual information at different resolutions and a cascaded feature attention mechanism to refine spatial and channel-wise features. The proposed framework aims to improve feature representation and localization for more accurate liver segmentation.

Background

Medical image segmentation is essential for applications such as tumor detection, organ volume estimation, radiotherapy planning, and disease monitoring. CT scans are widely used for liver imaging due to their clinical reliability. However, liver segmentation is challenging because of:

- Similar intensity between liver and surrounding organs
- Irregular tumor shapes and size variations
- High inter-patient variability

- Noise and imaging artifacts
- Class imbalance (tumor regions are very small)

Traditional techniques such as thresholding, edge detection, and region growing often fail in complex cases due to limited contextual understanding. Deep learning models, particularly U-Net and its variants, address these limitations by enabling:

- Pixel-level prediction
- End-to-end learning
- Robust feature extraction
- Multiscale contextual understanding .

Objective

The primary objective of this project is to develop an automated deep learning-based framework for accurate liver and tumor segmentation from CT scans using the LiTS dataset. Specific objectives include:

- Preprocessing 3D CT volumes, including normalization and slice preparation for effective model training
- Addressing class imbalance between background, liver, and tumor regions
- Designing and training a U-Net-based segmentation model for precise pixel-level prediction
- Evaluating model performance using metrics such as Dice Score and Accuracy
- Performing qualitative visual analysis to assess segmentation quality and boundary precision

Problem Statement

Liver tumor segmentation from CT scans is a complex and challenging task due to several factors, including low contrast between the liver and adjacent tissues, significant inter-patient anatomical variability, and the presence of extremely small tumor regions. Additionally, the 3D volumetric nature of CT data increases computational complexity and makes accurate boundary delineation more difficult.

The core problem addressed in this project is:

How can a deep learning-based segmentation framework accurately identify and segment liver and tumor regions from 3D CT scans while effectively managing class imbalance, preserving fine structural details, and maintaining computational efficiency?

Dataset

The dataset used in this project is the Liver Tumor Segmentation (LiTS) dataset, publicly available on Kaggle.The dataset contains abdominal CT scans along with corresponding ground truth segmentation masks.

- **Part 1:**
  `https://www.kaggle.com/datasets/andrewmvd/liver-tumor-segmentation`
- **Part 2:**
  `https://www.kaggle.com/datasets/andrewmvd/liver-tumor-segmentation-part-2`

Both parts together form the complete dataset.

0.1. File Structure

The dataset is organized into CT volumes and corresponding segmentation masks stored in NIfTI (.nii) format. A typical directory structure is as follows:

```
liver-tumor-segmentation/
|
|-- volume-0.nii
|-- volume-1.nii
|-- ...
|
```

```
|-- segmentation-0.nii
|-- segmentation-1.nii
|-- ...
```

Each case contains:

- **volume-*.nii** — 3D CT scan of a patient
- **segmentation-*.nii** — Ground truth segmentation mask of the same dimensions

0.2. Data Characteristics

The dataset consists of 3D volumetric CT scans where each volume is composed of multiple 2D slices stacked along the z-axis.

The segmentation masks contain three classes:

- 0 — Background
- 1 — Liver
- 2 — Tumor

Each segmentation file corresponds exactly to its respective CT volume, ensuring spatial alignment between image data and annotations.

The dataset reflects real clinical variability in liver morphology, tumor size, tumor location, and intensity distribution in different patients, making the segmentation task both realistic and challenging.

Related Research Papers

Several deep learning-based approaches have been proposed for liver and liver-tumor segmentation in CT images, focusing on improving segmentation performance.

Xie et al. (2021) proposed a Dynamic Adaptive Residual Network for liver CT image segmentation. Their approach leverages adaptive residual learning to improve feature propagation and enhance segmentation accuracy, particularly in complex anatomical regions.

Seo et al. (2020) introduced a Modified U-Net (MU-Net) that incorporates object-dependent high-level features for improved liver and liver-tumor segmentation. By integrating semantic-level contextual information, their

method enhances discrimination between liver tissues and surrounding organs.

Jiang et al. (2019) developed AHCNet, which applies an attention mechanism combined with hybrid connections for liver tumor segmentation in CT volumes. The attention mechanism enables the network to focus on relevant spatial regions, thereby improving tumor localization performance.

Data Cleaning

The following data cleaning and preprocessing steps were performed on the LiTS dataset to ensure data consistency, reduce noise, and improve model performance:

- **Volume–Mask Verification:** Mismatched CT volumes and segmentation masks were removed by verifying file correspondence (e.g., replacing "volume" with "segmentation" in filenames) to ensure correct volume–mask pairing.
- **Reorientation of Volumes:** CT volumes and masks were reoriented from shape (H, W, D) to (D, H, W) to maintain a consistent depth-first format required for 3D convolutional processing.
- **Hounsfield Unit (HU) Clipping:** Intensity values were clipped to the range $[-200, 250]$ Hounsfield Units to suppress irrelevant anatomical structures while preserving liver and tumor regions.
- **Intensity Normalization:** Z-score normalization (mean subtraction followed by division by standard deviation) was applied to reduce inter-scan intensity variations and improve model convergence.
- **Binary Mask Conversion:** Segmentation masks were converted into binary format by mapping the tumor label (2) to 1 and treating all other labels as background.
- **Removal of Non-Informative Slices:** Slices containing only background pixels were removed to reduce class imbalance and eliminate unnecessary computation.
- **3D Patch Extraction:** Volumetric patches of size $(16 \times 128 \times 128)$ were extracted using a fixed stride to enable efficient 3D training.

- **Tumor-Focused Patch Selection:** Only patches containing tumor pixels were retained to further address tumor–background imbalance and focus training on informative regions.
- **Dataset Subsampling:** A reduced subset of the dataset was used during initial experimentation due to computational and GPU memory constraints.

To further mitigate class imbalance during training, patch-based sampling was combined with a Dice-based loss function, which emphasizes overlap between predicted and ground truth regions.

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the statistical characteristics and structural variability of the LiTS dataset prior to model training. The following key observations were obtained:

- **Number of CT Volumes:** The total number of volumetric CT scans used in the study was computed to assess overall dataset scale and diversity.
- **Average Number of Slices per Volume:** CT volumes exhibited variability in depth. The average number of slices per volume was calculated to evaluate dimensional heterogeneity across different patients.
- **Intensity Distribution Analysis:** Statistical analysis of Hounsfield Unit (HU) values revealed a wide range of intensity variations across scans. The computed mean and standard deviation confirmed the necessity of HU clipping and z-score normalization to reduce inter-scan variability and stabilize training.
- **Distribution of Liver Area:** The liver occupies a significant yet variable portion of each slice. Pixel-level analysis demonstrated considerable variation in liver area across patients due to anatomical differences.
- **Distribution of Tumor Area:** Tumor regions were substantially smaller and highly inconsistent in size across volumes. This variability further increases segmentation complexity.

- **Tumor-to-Total Pixel Ratio:** Tumor pixels constitute a very small fraction of the total voxels in the dataset, indicating severe class imbalance. This observation justifies the adoption of patch-based sampling strategies and Dice-based loss functions during training.

Overall, the EDA findings highlight substantial inter-patient variability and pronounced foreground–background imbalance, reinforcing the need for robust multiscale feature extraction and attention mechanisms in the proposed segmentation framework.

Data Visualization

Data visualization was performed to qualitatively analyze the segmentation masks and understand spatial relationships between anatomical structures.

Ground Truth Mask Overlay

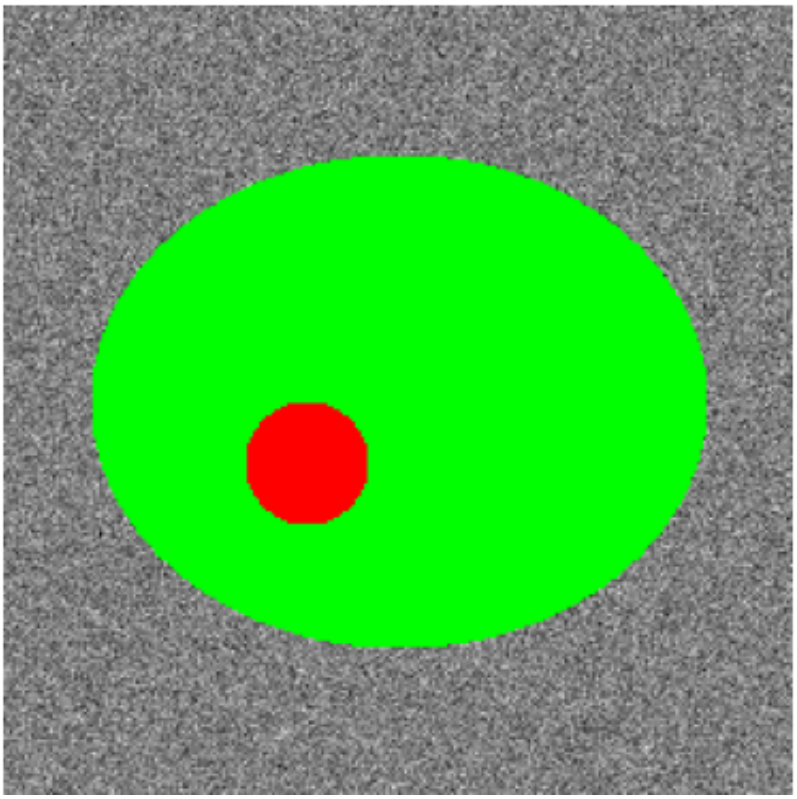

Figure 1: Ground truth mask overlay on CT image. Background regions are shown in black, liver regions in green, and tumor regions in red.

Pixel Distribution Bar Chart

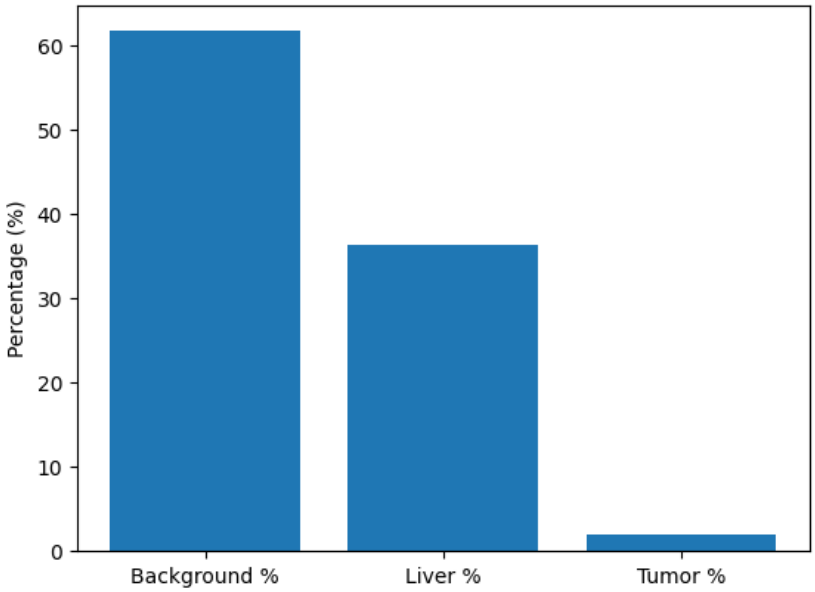

Figure 2: Pixel distribution bar chart showing class imbalance in the dataset. Background constitute majority pixels while tumor pixels constitute minority of the total voxels.

References

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